

A): FIRST FIT

C PROGRAM:

```
#include <stdio.h>

int main()
{
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb, np, i, j;

    printf("Enter Number of Blocks: ");
    scanf("%d",&nb);

    printf("Enter Number of Processes: ");
    scanf("%d",&np);

    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d",&blockSize[i]);

    printf("Enter Process Sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d",&processSize[i]);

    for(i=0;i<np;i++)
    {
        allocation[i]=-1;
        for(j=0;j<nb;j++)
        {
            if(blockSize[j] >= processSize[i])
            {
                allocation[i]=j;
```

```

blockSize[j]-=processSize[i];
break;
}
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
else
printf("Not Allocated\n");
}
return 0;
}

```

SHELL SCRIPTING:

```

#!/bin/bash

echo "First Fit Memory Allocation Demonstration"

blocks=(100 500 200 300 600)

processes=(212 417 112 426)

echo "Memory Blocks: ${blocks[@]}"

echo "Processes: ${processes[@]}"

echo "Allocation Performed Using First Fit"

```

OUTPUT:

```
(kali㉿Anbuchelvan)-[~]  
$ nano firstfit.sh  
  
(kali㉿Anbuchelvan)-[~]  
$ chmod +x firstfit.sh  
  
(kali㉿Anbuchelvan)-[~]  
$ bash firstfit.sh  
First Fit Memory Allocation Demonstration  
Memory Blocks: 100 500 200 300 600  
Processes: 212 417 112 426  
Allocation Performed Using First Fit
```

B: BEST FIT

C PROGRAM:

```
#include <stdio.h>  
  
int main()  
{  
    int blockSize[20], processSize[20];  
    int allocation[20];  
    int nb,np,i,j,bestIdx;  
    printf("Enter Number of Blocks: ");  
    scanf("%d",&nb);  
    printf("Enter Number of Processes: ");  
    scanf("%d",&np);  
    printf("Enter Block Sizes:\n");  
    for(i=0;i<nb;i++)  
        scanf("%d",&blockSize[i]);  
    printf("Enter Process Sizes:\n");  
    for(i=0;i<np;i++)  
        scanf("%d",&processSize[i]);
```

```

for(i=0;i<np;i++)
allocation[i]=-1;
for(i=0;i<np;i++)
{
bestIdx=-1;
for(j=0;j<nb;j++)
{
if(blockSize[j] >= processSize[i])
{
if(bestIdx==-1 || blockSize[j] < blockSize[bestIdx])
bestIdx=j;
}
}
if(bestIdx!=-1)
{
allocation[i]=bestIdx;
blockSize[bestIdx]-=processSize[i];
}
}
printf("\nProcess No\tProcess Size\tBlock No\n");
for(i=0;i<np;i++)
{
printf("%d\t\t%d\t\t",i+1,processSize[i]);
if(allocation[i]!=-1)
printf("%d\n",allocation[i]+1);
else
printf("Not Allocated\n");
}

```

```
return 0;
```

```
}
```

SHELL SCRIPTING:

```
#!/bin/bash
```

```
echo "Best Fit Memory Allocation Demonstration"
```

```
blocks=(100 500 200 300 600)
```

```
processes=(212 417 112 426)
```

```
echo "Memory Blocks: ${blocks[@]}"
```

```
echo "Processes: ${processes[@]}"
```

```
echo "Allocation Performed Using Best Fit"
```

OUTPUT:

```
(kali㉿ Anbuchelvan)~  
$ nano bestfit.sh
```

```
(kali㉿ Anbuchelvan)~  
$ chmod +x bestfit.sh
```

```
(kali㉿ Anbuchelvan)~  
$ bash bestfit.sh
```

```
Best Fit Memory Allocation Demonstration
```

```
Memory Blocks: 100 500 200 300 600
```

```
Processes: 212 417 112 426
```

```
Allocation Performed Using Best Fit
```

C: WORST FIT

C PROGRAM:

```
#include <stdio.h>

int main()
{
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,worstIdx;
    printf("Enter Number of Blocks: ");
    scanf("%d",&nb);
    printf("Enter Number of Processes: ");
    scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d",&blockSize[i]);
    printf("Enter Process Sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d",&processSize[i]);
    for(i=0;i<np;i++)
        allocation[i]=-1;
    for(i=0;i<np;i++)
    {
        worstIdx=-1;
        for(j=0;j<nb;j++)
        {
            if(blockSize[j] >= processSize[i])
            {
                if(worstIdx==-1 || blockSize[j] > blockSize[worstIdx])
```

```

worstIdx=j;

}

}

if(worstIdx!=-1)

{

allocation[i]=worstIdx;

blockSize[worstIdx]-=processSize[i];

}

}

printf("\nProcess No\tProcess Size\tBlock No\n");

for(i=0;i<np;i++)

{

printf("%d\t\t%d\t\t",i+1,processSize[i]);

if(allocation[i]!=-1)

printf("%d\n",allocation[i]+1);

else

printf("Not Allocated\n");

}

return 0;

}

```

SHELL SCRIPTING:

```
#!/bin/bash
```

```
echo "Worst Fit Memory Allocation Demonstration"
```

```
blocks=(100 500 200 300 600)
```

```
processes=(212 417 112 426)
```

```
echo "Memory Blocks: ${blocks[@]}"
```

```
echo "Processes: ${processes[@]}"
```

```
echo "Allocation Performed Using Worst Fit"
```

OUTPUT:

```
(kali㉿ Anbuchelvan)-[~]  
$ nano worstfit.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ chmod +x worstfit.sh
```

```
(kali㉿ Anbuchelvan)-[~]  
$ bash worstfit.sh
```

Worst Fit Memory Allocation Demonstration

Memory Blocks: 100 500 200 300 600

Processes: 212 417 112 426

Allocation Performed Using Worst Fit